

AI-Driven Health Intelligence System

Revolutionizing Healthcare Delivery through Advanced Artificial Intelligence and Seamless Medical Device Interoperability.

 Salvatory Marko Sanga — Data Thinker & System Builder

| The Problem



Severe Bottlenecks in Patient Flow

Valuable clinical hours are lost as patients navigate between primary physicians, diagnostic labs, and dedicated imaging suites (such as CT-scans).



Highly Siloed Diagnostic Hardware

No cohesive pipeline exists to directly link raw medical device telemetric output with patient records, leading to manual errors and lags.



The Rural Diagnostic Specialist Divide

Rural and primary healthcare centers completely lack on-premise expert radiologists to quickly interpret critical diagnostic scans.

| Our Solution

Unified Health Intelligence

An innovative cloud-scale health platform integrating comprehensive patient electronic records, direct hardware communication channels, and real-time diagnostic AI models.

This approach eliminates communication barriers across hospital units, automates medical telemetry, and instantly structures crucial clinical workflows for fast-track interventions.



| AI Diagnostics

Automated Scans Analysis

Clinicians easily upload diagnostic images directly for instant AI analysis.

The neural engine returns reliable confidence scores (%), identifies pathological states, and flags warning indicators to assist medical staff.



| IoT & Device Telemetry

Direct Hardware Integration

Supports native communication interfaces with specialized imaging platforms (CT/MRI/X-Ray) and medical labs through automated cloud channels.

Zero Human-Input Friction

Data streams securely bypass manual record logging entirely. Eliminates human data-entry mistakes while delivering immediate diagnostic readouts.

| Rural-to-Referral Network

Decentralizing Expertise

Allows local practitioners to scan and upload clinical materials securely, instantly projecting state-of-the-art diagnostic resources to remote centers.

Major urban referral center specialists can remotely validate diagnostic results and oversee local care pipelines, optimizing specialized coverage.



Machine Learning Architecture



Triage & Diagnostics

Deep models prioritizing incoming workflows based on the critical urgency of patients' conditions.



Pregnancy Risk AI

Specialized maternal predictive algorithms identifying potential clinical risks well in advance.



Equipment Guard

Predictive hardware maintenance models logging telemetry to ensure clinical scanners operate error-free.

| Market Opportunity

Clinical System Demands

Public and private clinical operators are under continuous pressure to handle growing patient volumes with limited administrative and specialist staff resources.

Scale & Interoperability

Growing regional demand for secure, cloud-enabled hospital middleware that connects disconnected on-site medical devices to central records.

Competitive Advantage

Capability Feature	Standard HMS Systems	Our Health Intelligence
AI Diagnostics	None / Pure Manual Assessment	Direct Computer Vision Engines
Device Connection	Manual CSV & Paper Imports	Instant Telemetry Integration
Infrastructure Support	Basic Patient Database Records	4 Active Predictive ML Models
Scalability Design	Highly Rigid / Local Silos	150+ Enterprise API Endpoints

Architecture & Scalability

150+

Active API Endpoints

18

Integrated UI Pages

Production-Ready Foundation

Engineered from the ground up for extreme data loads. Our back-end services process parallel device inputs, clinical triage models, and custom image ingestion models with high stability.

The architecture ensures secure, role-based authorization for physicians, administrators, and lab technologists.

| The Team



Salvatory Marko Sanga

System Architect & Software Builder

Pursuing Computer Science credentials at **Mbeya University of Science and Technology (MUST)**. Deeply focused on back-end architectures, data engineering pipelines, and clinical machine learning models.

Independently developed this end-to-end platform architecture, managing the core database designs and multi-model backend integration.

Development Roadmap



Phase 1

Core Backend Completed

150+ Endpoints built. Native ML model engines deployed.



Phase 2

UI/UX Refinement

Transitioning functional interfaces to full Enterprise UI dashboard.



Phase 3

Clinical Pilots

Deploying trial runs in active health facilities for validation.

Image Sources



[https://images.philips.com/is/image/PhilipsConsumer/a98deec23a0349ed9717b10e00213547?extend=0,-305,0,-535&wid=1922&hei=641&fit=stretch,1&\\$jpglarge\\$](https://images.philips.com/is/image/PhilipsConsumer/a98deec23a0349ed9717b10e00213547?extend=0,-305,0,-535&wid=1922&hei=641&fit=stretch,1&$jpglarge$)

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